ANTONIO ARBIZZANI TONELLI · EPFL MACHINE LEARNING CS-433

Cardiovascular disease classification in NumPy

Codebook cleaning, diet representation and restricted imputation, followed by histogram gradient boosting written in NumPy.

AIcrowd F1
0.455
Reported challenge feedback

AIcrowd accuracy
0.882
Reported challenge feedback

AIcrowd leaderboard
Ranked 3rd
Leaderboard snapshot: 14 October 2026

Position reported by the author; challenge scores from recorded author-reported AIcrowd feedback · AIcrowd leaderboard

One inspectable pipeline produces the result

1
Codebook cleaning
Survey codes by meaning

2
Diet representation
Keep per-day fields

3
Restricted imputation
Fitted on training rows

4
NumPy boosting
200 trees, depth 5

5
OOF threshold
Calibrated on all rows

6
Submission
109,379 predictions

RESULT

F1 0.455

Reported AIcrowd feedback, accuracy 0.882. Ranked 3rd on 14 October 2026 (author-reported).

LOCAL EVIDENCE

+0.0077 F1

Historical booster vs matched logistic regression on the same rows, cleaning and nested folds.

LIMITS

Reported disease

No nested estimate exists for the final combination. HAREHAB1 holds post-event information.

Model and threshold are fixed for the 14 October 2026 release. Challenge feedback informed the final choice.

Source: project/configs/final_model.json, evidence/RELEASE_20261014.json

A rare positive class and heavy missingness

328,135
labeled respondents;
109,379 challenge test rows

8.83%
positive labels (28,975 respondents)

44.79%
of raw predictor cells are missing

321
raw predictors in the survey extract

Why F1 guides selection

Predicting every respondent as negative gives 91.17% accuracy and F1 0: no positive is found.



8.83% positive **91.17% negative**

On labeled data, an all-negative baseline gives 91.17% accuracy and F1 0. AIcrowd reports F1 0.455 and accuracy 0.882.

Target **_MICHD**: reported myocardial infarction or coronary heart disease in BRFSS 2015. A cross-sectional classification task, not prospective risk prediction.

Source: evidence/METRICS.json, project/docs/DATA.md

Survey codes are interpreted before any fit

Survey code

`ALCDAY5 = 888`

Recoded as

0

No alcohol consumption means zero drinking days.

Survey code

`PHYSHLTH = 88`

Recoded as

0

Zero unhealthy days is a real quantity, not a category.

Survey code

`Nonresponse codes`

Recoded as

missing

Each variable's own unknown and refused codes, not a global 7/9 rule.

1,095,421
nonresponse codes set to missing

899,687
zero-quantity codes set to 0

Counts from the historical audit of 262,508 development rows. They describe the transformation, not a performance gain.

Source: configs/experiments/ (codebook plan), results/eda/ (preprocessing counts)

The final representation adds two targeted changes

Diet representation

Remove six raw diet questions that mix day, week and month codes; keep the existing derived per-day fields.

`FRUIT1 → FRUTDA1_ (kept)`

Restricted imputation

Fill only selected unknown or refused answers (income, employment, pneumonia vaccine) and blank height and weight from training features; derive missing BMI.

297 model features

289 source columns + 8 missing-state flags

Structural missingness is preserved. Imputers use features, never disease labels, and are fitted inside each training partition.

Both changes are in the final pipeline because of challenge feedback; under local nested validation they were neutral (see the evidence section).

Source: configs/experiments/combined18_24_preprocessing.json, src/conditional_imputation.py

Histogram gradient boosting, implemented in NumPy

1 Bin

64 quantile bins per feature, learned on training rows

2 Fit

Each tree takes a Newton step on the logistic loss, depth up to 5

3 Sum

200 tree outputs are added with learning rate 0.05

4 Decide

Probability above the calibrated threshold 0.2181 is positive

MISSING VALUES

Each split tests both directions for missing responses, so no global constant replaces them.

CONFIGURATION

200 trees · depth 5 · 128 candidate features per node · at least 200 rows per leaf · leaf L2 1

Training uses NumPy and the Python standard library. The booster source is the historical implementation, kept byte-identical so the release replays exactly.

Source: src/numpy_boosting_phase14.py, configs/final_model.json

Each reported score answers a different question

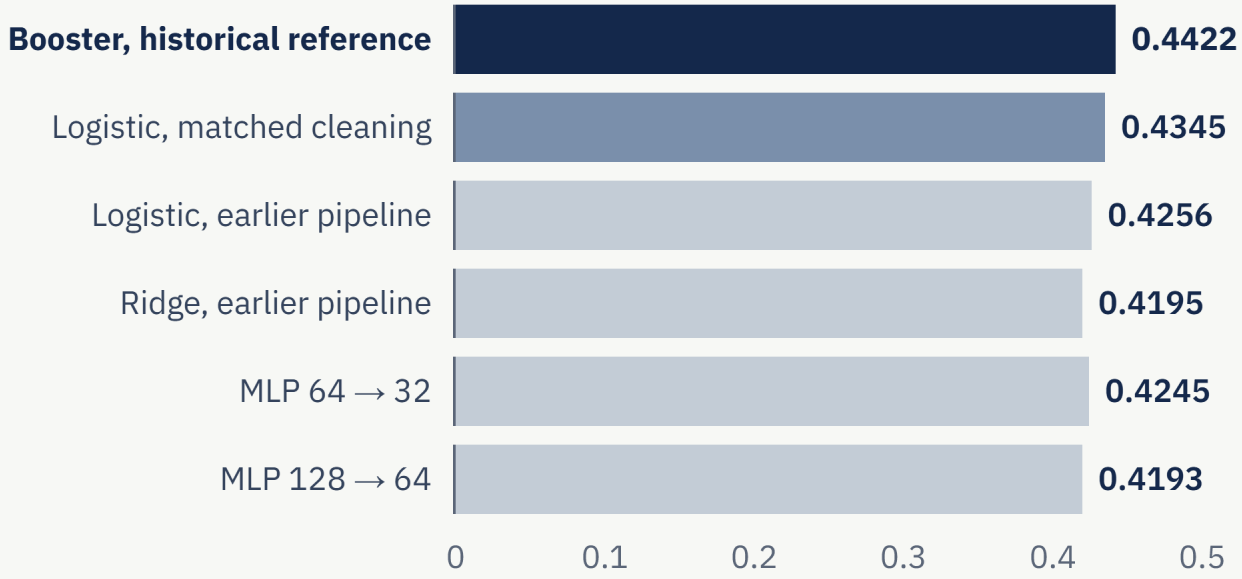
0.4422	**Historical nested development** Historical reference booster, scored on excluded outer folds of 262,508 rows.
0.4439	**Full-train OOF calibration · selected pipeline** Picks the threshold on all 328,135 rows; the same labels score it: not a test.
0.455	**AIcrowd feedback · selected pipeline** Reported challenge score, three decimals; it informed the final choice.
n/a	**Nested estimate · selected pipeline** Not recorded; historical nested scores are not reassigned to it.

These are not successive improvements: the numbers come from different protocols, populations and pipelines.

Source: evidence/METRICS.json, results/eda/final_combined18_24_submission.json

Locally, the booster beat a matched linear baseline

Historical nested development F1 (axis 0 to 0.50)



+0.0077

F1, booster minus matched logistic

95% paired interval [+0.0049, +0.0105]; higher in all three outer folds

Shared cleaning, 295 source variables, rows and folds; each model keeps its own matrix.

Historical comparison of complete pipelines. It supports the choice of model family; it is not a held-out score for the final combination.

Source: assets/chart_data/model_comparison.csv, results/eda/phase38_matched_logistic_summary.md

No tested change beat the historical reference locally

Tested change	Nested F1	vs 0.4422	Outcome
MLP 128 → 64	0.4193	−0.0229	Not adopted
MLP 64 → 32	0.4245	−0.0177	Not adopted
Random undersampling 5:1	0.4382	−0.0039	Not adopted
Top-100 features	0.4392	−0.0029	Not adopted
Positive weight 1.25	0.4408	−0.0013	Not adopted
Restricted imputation	0.4418	−0.0004	In final pipeline
Diet representation	0.4419	−0.0003	In final pipeline

Two neutral changes made the final cut

Locally both were within 0.0004 F1 of the reference. They entered the final pipeline because their full-train submissions scored higher on AIcrowd.

A challenge decision, not proof that they generalize better.

Historical nested development on the same 262,508 rows. These results guide choices among the tested settings; they do not rule out better tuning of MLPs, weighting or feature selection.

Source: evidence/METRICS.json, project/docs/MLP.md, project/docs/EXPERIMENTS.md

Semantic cleaning is justified by meaning, not by a large gain

Raw predictors (321)

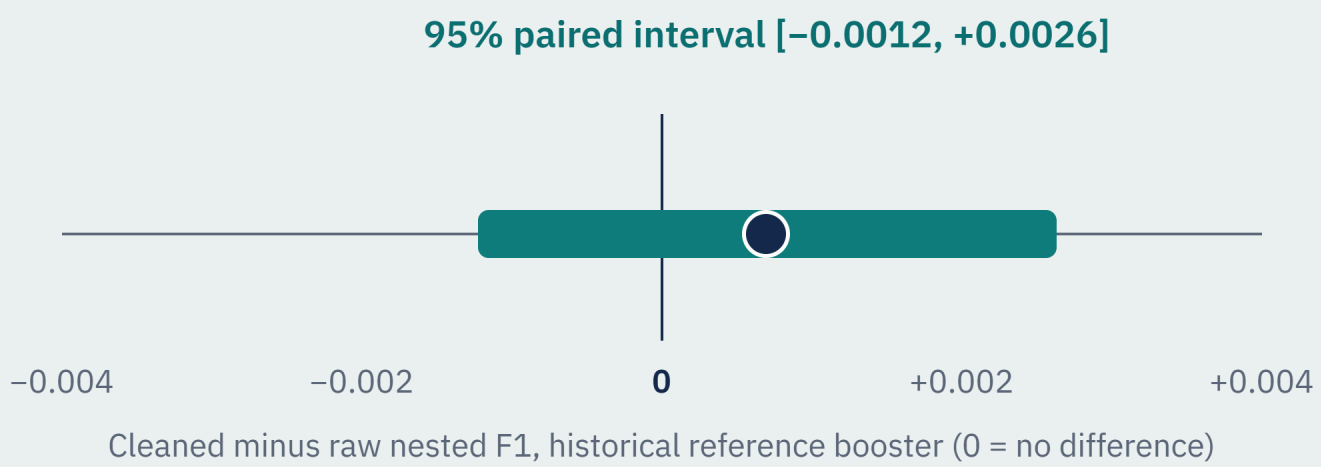
0.4415

Cleaned predictors (295)

0.4422

Difference in nested F1

+0.0007



The interval includes zero. The benchmark measures recoding and feature exclusions together and is conditional on fixed fits and thresholds.

Source: results/eda/phase33_raw_preprocessing_summary.md, evidence/METRICS.json

The selected combination has the highest reported challenge F1

Submitted pipeline	F1	Accuracy
Historical reference booster	0.452	0.878
+ Diet representation	0.453	0.882
+ Restricted imputation	0.454	0.880
Leaf L2 = 5	0.453	0.883
Diet + restricted imputation, L2 1 (selected)	0.455	0.882
Diet + restricted imputation, L2 5	0.453	0.883

+0.003

F1 above the historical submission

Six submissions span 0.452 to 0.455, the third decimal of a rounded display: no significance is claimed. Accuracy did not decide.

Author-reported AIcrowd display scores: historical reference on 12 October, other candidates on 13 October 2026. Rank 3rd on 14 October 2026 is reported separately by the author.

Source: assets/chart_data/aicrowd_progression.csv, evidence/aicrowd/

Historical error analysis shows the F1 trade-off

	Predicted positive	Predicted negative
Actual positive 23,180	**12,855** True positives (found)	**10,325** False negatives (missed)
Actual negative 239,328	**22,112** False positives	**217,216** True negatives

0.3676

Precision: of 34,967 flagged respondents, about 37% are positive

0.5546

Recall: 12,855 of 23,180 positives are found

Historical reference booster, nested development decisions. No equivalent confusion matrix exists for the final combination; its 13,999 positive test predictions are counts, not errors.

Source: evidence/METRICS.json, evidence/SELECTED_REPLAY_20261014.json

A post-event question limits what the result can claim

Heart attack
The event comes first.

→

HAREHAB1
Asks about rehabilitation after a heart attack.

→

_MICHD
The target: reported heart attack or coronary disease.

620 / 620
recorded answers belong to positives; only 0.24% of development rows have one

0.4422 → 0.4310
historical nested F1 with and without the feature

The selected pipeline keeps HAREHAB1. It classifies reported disease and does not establish prediction before an event; a prospective study would need a new feature policy and fresh validation.

Source: evidence/METRICS.json, project/docs/FINAL_REPORT.md

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Results, code and evidence in one repository

Selected NumPy pipeline · reported AIcrowd F1 0.455 · ranked 3rd on 14 October 2026, as reported by the author.

VERIFICATION

242 synthetic tests pass locally and in a clean package; the selected predictions replay exactly.

START HERE

`docs/RESULTS.md`
`project/README.md`
`docs/CODE_REVIEW.md`
`evidence/`

PROPOSED NEXT STEP

A nested estimate for the final pipeline, then a version without HAREHAB1 scored once on untouched data.

Code, configurations, results and limitations

github.com/antonio-arbizzani-tonelli/cardiovascular-classification-numpy

Appendix A · Selected release specification

Setting	Selected value
Pipeline identifier	combined18_24_l2_1
Repository phases	Phase 18 + Phase 24 + Phase 14 booster
Model input	289 source columns + 8 flags = 297
Trees · depth · learning rate	200 · 5 · 0.05
Leaf rows · bins · features	≥ 200 · 64 · 128 per node
Leaf L2 · seed	1 · 20260920
Decision threshold	0.21811010373987366
Threshold calibration	3-fold full-train OOF, seed 20260919

Reproduce from project/

```
pip install -r requirements-dev.txt
python -m tools.verify_repository
python -m unittest discover
-s tests -q
python run.py
--data-dir data/raw/dataset
--output submission.csv
```

Training needs the official challenge CSVs. Data, checkpoints and row-level predictions stay outside Git.

The booster source is preserved byte-identical for replay; preprocessing and threshold identify the new pipeline.

Source: project/configs/final_model.json, project/README.md

Appendix B · Metrics and release scores

Historical pipeline (nested)	F1	Precision	Recall	AP
Ridge, earlier pipeline	0.4195	0.3330	0.5667	0.3903
Logistic, earlier pipeline	0.4256	0.3573	0.5263	0.3877
Logistic, matched cleaning	0.4345	0.3675	0.5312	0.4210
Booster, historical reference	0.4422	0.3676	0.5546	0.4286
MLP 64 → 32	0.4245	0.3607	0.5155	0.3878
MLP 128 → 64	0.4193	0.3400	0.5467	0.3794

Selected release (combined pipeline): OOF calibration F1 0.4439 · AIcrowd F1 0.455, accuracy 0.882 · ranked 3rd on 14 October 2026 (author-reported, not independently verified).

No nested held-out estimate is recorded for the combined pipeline, so it is not listed in the table.

Nested development: 262,508 rows, 3 outer / 2 inner folds. AP = average precision. Source: evidence/METRICS.json